Automating the Screening Process of Profitable Stocks with Highest 3 Years Average Annual Returns

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1. Context

This project is in the domain of Financial Technology (FinTech), targeted towards investors, portfolio managers, and financial analysts. The total number of stocks listed on the Indian exchange makes manual analysis really very inefficient. By automating the process of screening stocks that are profitable based on the historical trends especially the close price and monthly returns, the project aligns with recent industry trends of using AI and data analytics for investment decisions.

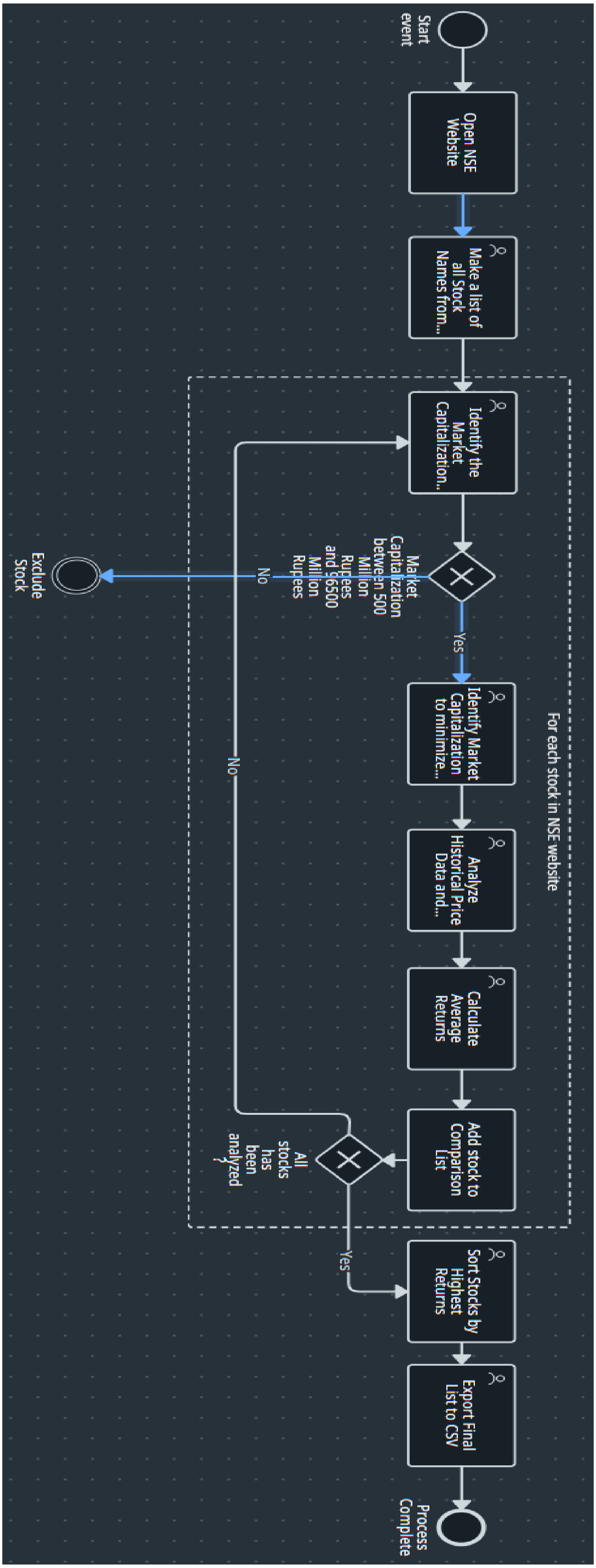
The automated process will streamline the stock information collection process and removes manual calculations that are redundant. By applying filters on market capitalization and profitability levels, the system shortlists the high-flying stocks only. The entire filtering parameters are encoded within the logic so that consistency and reproducibility are assured.

Screening stocks for profitability using fundamental and technical parameters is crucial for investors but is very manual, repetitive, and data-intensive. By automating the identification of Indian stocks listed on NSE stock exchange with the highest 3-year average yearly returns, it is possible enable easier and quicker decision-making for investment portfolios. The automation approach mainly uses the YFinance APIs and Python scripting for fetching the data, categorization and classification, profitability analysis, and calculation of the highest returns to identify profitable stocks.

2. As-Is

The process begins manually with the scraping of stock names from the NSE website [1]. Then, market capitalization data must be searched from financial sites or reports. Analysts use Excel to gather 3-year monthly price data and perform manual growth percentages or sort stocks by size. Validation is not performed to test integrity of the data or analysis of performance. This is performed on a regular basis, e.g., monthly or quarterly.

See attached BPMN diagram showing the current manual process [5][6].


Start event
Open NSE Website
Make a list of all Stock Names from...
For each stock in NSE website
Identify the Market Capitalization...
Market Capitalization between 500 Million Rupees and 96500 Million Rupees
No
Exclude Stock
Yes
Identify Market Capitalization to minimize...
Analyze Historical Price Data and...
Calculate Average Returns
Add stock to Comparison List
All stocks has been analyzed ?
No
Yes
Sort Stocks by Highest Returns
Export Final List to CSV
Process Complete

3. As-Is Analysis

The As-Is process analysis is purely manual, thereby utilizing human resources and efforts that can be otherwise utilised else in efficient decision-making tasks. It also comes with increased risk of errors and bias. The process varies and is affected by the knowledge capacity of different human resource.

3.1. Manual Processes

- Gathering data from different sources.
- Manual input and organization of data.
- Trend analysis based on visual observation.
- Computation of returns using use of spreadsheets.
- Aggregation and sorting of results.

3.2. Cost Impacts

Manual research carries a very high time expense for financial analysts. Simple screening of 100 stocks will take hours. The process is not appropriate in real-time or repetitive execution of tasks because human effort is very inefficient for real-time or repetitive execution of tasks. Opportunity cost of delayed investment choice. Potential loss due to miscalculation.

3.3. Error Rates

Human judgment and clerical errors increase the likelihood of mistake in determining the real returns and trends. Inconsistent formats from data sources will contribute to mistakes and intensify it. There will be approximately 5-7% rate of error with manual computation, 10-15% discrepancy in the analysis of trends and repeated data entry mistakes.

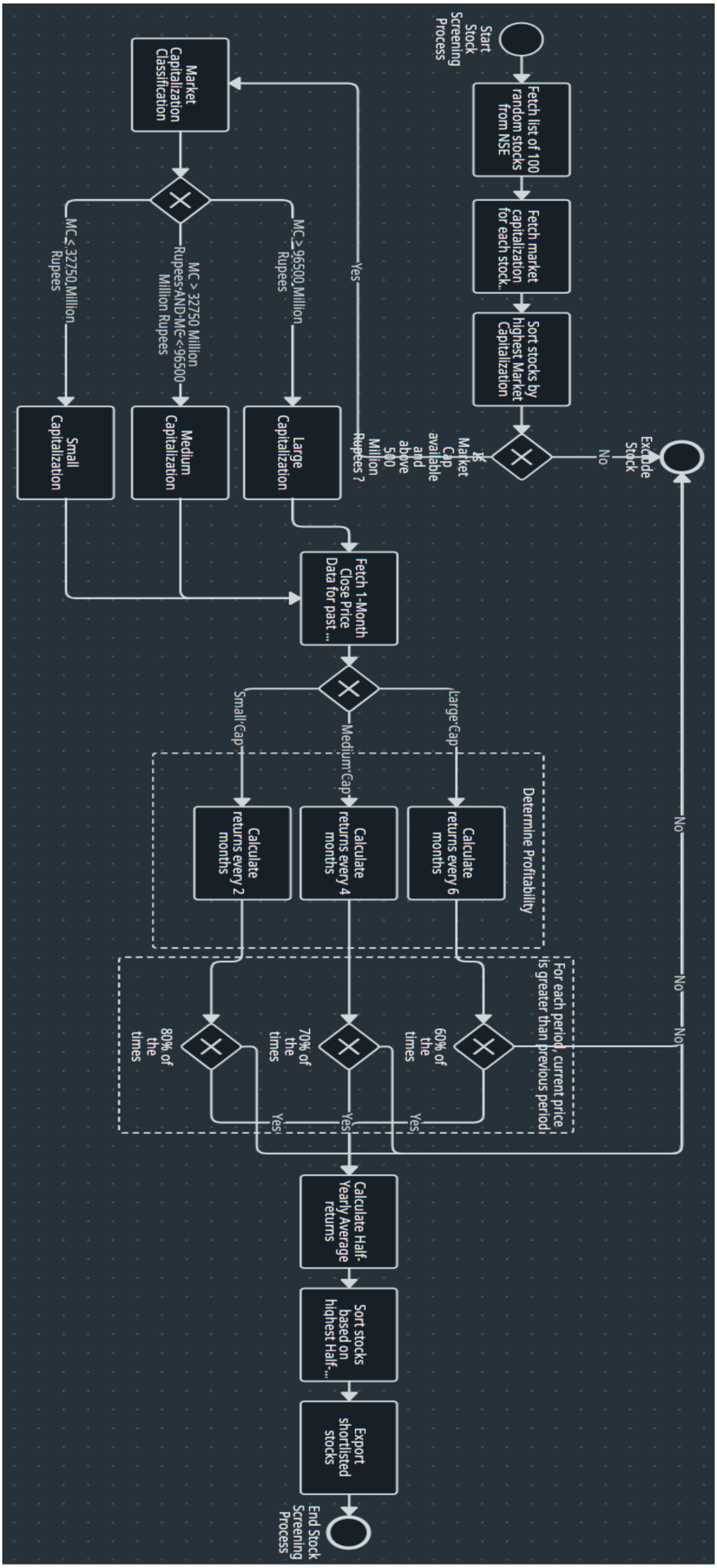
3.4. Lost Productivity

Frequent data lookups and computations as a repeated task detract time from high-value analysis, reducing productivity in financial decision-making roles. 80% of analysis time spent in data gathering or processing. This will result in low capacity for thorough analysis typically resulting in less than100 stocks to be screened. Not being able to re-run analyses easily with changed parameters is another serious impact.

4. To-Be

The process begins by selecting a number of stocks from a stored list. YFinance API [2] are utilized to fetch and extract data directly into the Python scripts [4]. The stocks are sorted by market cap, sectioned by cap size, and screened for profitability in accordance with defined logic. The system tracks price increase frequency and calculates average bi-annual returns, outputting the best prospects ultimately into a ready CSV structure for research or investment intentions.

Please see attached BPMN diagram showing the automated process [5][6].


Start Stock Screening Process
Fetch list of 100 random stocks from NSE
Fetch market capitalization for each stock.
Sort stocks by highest Market Capitalization
Is Market Cap available and above 500 Million Rupees ?
No
Exclude Stock
Yes
Market Capitalization Classification
MC ≥ 96500 Million Rupees
MC > 32750 Million Rupees AND MC < 96500 Million Rupees
MC ≤ 32750 Million Rupees
Large Capitalization
Medium Capitalization
Small Capitalization
Fetch 1-Month Close Price Data for past...
Large-Cap
Medium-Cap
Small-Cap
Determine Profitability
Calculate returns every 6 months
Calculate returns every 4 months
Calculate returns every 2 months
For each period, current price is greater than previous period
60% of the times
70% of the times
80% of the times
Yes
Yes
Yes
No
No
No
Calculate Half-Yearly Average returns
Sort stocks based on highest Half-...
Export shortlisted stocks
End Stock Screening Process

5. To-Be Analysis

5.1. Manual Processes Eliminated

• Automated the data fetching, collection, preprocessing and filtering steps.
• Analysis of stock data and trends has been done programmatically.
• Calculations of average annual returns has been automated and standardized.
• The final list of profitable stocks has been exported to file automatically.

5.2. Cost Savings

Automation has eliminated over 90% of the analysis time, thereby allowing real-time stock screening at a large scale with minimal human involvement and errors. Automation is thus an initial one time expense with very little maintenance. Processing time has been reduced from 8-10 hours to less than 5 minutes. Estimated cost savings of the analysis can be reduced to around 99% by using automation. Automation will also enable daily screening of the stock data for very minimal or decreased marginal cost.

5.3. Error Rates Reduction

Data are directly fetched from the financial YFinance APIs, which will reduce transcription and computational errors. It will also prevent loss of data by human errors. Conditions for filtering and screening of stocks are applied consistently via rules and thresholds. Calculation errors reduced to almost to near-zero. Review of the stock performance is standardized. Validations tests are run automatically without any human intervention.

5.4. Productivity Gains

Automation of the screening of stocks will releases the financial analysts to focus more on investment strategy and decision making, portfolio planning, and other high-impact activities rather than the continuous repetitive tasks. Capacity to screen number of stocks at minimal time can be drastically increased from 100 to 1,000+ stocks which will allow to run multiple scenarios quickly.

6. Automation Recommendation

The selected Python script screens the stocks automatically 95% of the time. The human effort is only in typing in the list of stock symbols which can also be automated on the basis of specific needs. The code is reusable and could be scheduled or embedded into dashboards or UiPath workflows [6]. Smart automation is possible using Machine Learning to generate future price predictions using ARIMA, LSTM, or hybrid models, Natural Language Processing (NLP) for sentiment analysis of news on shortlisted stocks, Cognitive Automation for allowing the script to make thresholds programmable based on market conditions or risk tolerance preferences on the portfolio. These would bring predictive power and context-awareness. The stock screening feature is highly suitable for automation. Python [4] is mainly used because it is well established and contains abundance of libraries and frameworks such as YFinance and Pandas [3] which are easy to use. The future work may include adding ML-based trend forecast or NLP-based sentiment analysis.

7. Solution Demonstration

1. Fetch a list of 100 stocks from NSE exchange [1].
2. Fetch market capitalization for each stock.
3. Sort stocks by highest Market Capitalization.
4. If the Market Capitalization is not available and is not above 500 million rupees, exclude the stock. If the condition satisfies, proceed to Market Capitalization classification.
5. If the Market Capitalization is greater than 96500 million rupees, classify it as large company, if the Market Capitalization is between 32750 and 96500 million rupees, classify it as medium company, else classify it as small company.
6. Fetch 1-Month Close Price Data for past 3 years using YFinance API [2].
7. To determine profitability, for large company, calculate returns every 6 months, for medium company, calculate returns every 4 months and for small company, calculate returns every 2 months.
8. To reduce investment risk, filter the stocks based on the following:
   Exclude the stock if for each period, if the current price is not greater than previous period
   a. 60% of the times for large companies.
   b. 70% of the times for medium companies.
   c. 80% of the times for small companies.
9. Calculate Half-Annual Average returns of the past three years.
10. Sort stocks based on highest Half-Yearly Average returns.
11. Export shortlisted stocks to CSV file.

8. Conclusion

This project proves that even advanced data-driven processes like investment screening can be simplified using intelligent automation. The solution offers an effective solution for scalable, explainable, and adaptive automation techniques in the financial sector, especially in stock selection. The project also highlights the advantages and ease of use of Python can serve as an for automation solution development.

Automation of stock screening process has significantly enhanced the filtering pipeline for investments. With the help of python based financial APIs, historical data, and intelligent stock screening and filtering based on market capitalization and profitability trends, the automation solution helps in real-time fast screening of stocks that provide high-return opportunities for the analysts. The process is scalable and paves the way for future smart automation based on AI-powered forecasting.

8. References

[1] NSE India - https://www.nseindia.com/

[2] Yahoo Finance API (via YFinance) - https://pypi.org/project/yfinance/

[3] Pandas Documentation - https://pandas.pydata.org/

[4] Python Official Documentation - https://docs.python.org/3/

[5] Business Process Model and Notation (BPMN) - https://www.bpmn.org/

[6] UiPath Automation Framework - https://www.uipath.com/